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Robot-based Optical Assessment of Vital Signs in Healthcare with a Social Interface

Components, Workflows and Interfaces

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Abstract: Background Health kiosks with advanced imaging technologies can assess vital signs contactless. These kiosks, often found in supermarkets, pharmacies, and transit hubs, use cameras, thermal sensors, and AI to measure heart and respiratory rate, temperature, and oxygen saturation without physical contact. **Objective** We introduce a concept of an autonomous robotic system to scan the surroundings, detect humans, interact with them by leveraging a social interface, and gather their vital signs without physical contact. **Methods** A 6DOF robot arm is equipped with cameras to scan its surroundings. Upon detecting a person within the working space, the system prompts the person to step closer in a socially appropriate manner. Using contactless sensors, the system captures vital signs, such as the heart and respiratory rate, the body temperature and basic emotions by autonomously positioning the camera on the robotic arm to maintain a direct line of sight with the individual's face. **Results** The contribution provides the system architecture and first experiments with the sub-components. By integrating event-driven detection with unobtrusive monitoring, the proposed solution offers a novel robot-based method for routine, contactless vital sign collection, easing the workload of medical staff and patients.

Keywords: Healthcare robotics; Contactless vital signs, social user interaction, human-robot-interfaces, HRI

1 Introduction

The standardized recording and documentation of human vital

parameters (such as heart and respiratory rate, temperature, blood oxygen saturation (Spo2), blood pressure, weight, etc.) are tasks that are routinely carried out in doctors' offices, outpatient clinics, hospitals, nursing homes or emergency wards by trained medical staff. Except for acute emergency situations and intensive care units, where patients must be monitored continuously, these measurements are usually purely selective, and their interpretation only takes place within larger contexts in the form of multimodal longitudinal data.

In recent years, so-called 'health kiosks' [1] in various forms have been compiled and investigated as an alternative approach to record such vital data in a controlled and standardized environment from patients, home residents or even test subjects in preclinical studies. These *health kiosks* are booths in which vital signs can be recorded by the users themselves (or under telematic guidance) using cable-bound sensors, such as finger clips, blood pressure cuffs, etc., and – if required – support teleconsultation with clinical specialists [1]. Such health kiosks are promising, cost-efficient, and multi-functional devices and can potentially be integrated into existing clinical settings such as waiting areas [2], pharmacies, general practitioners or emergency facilities [3,4].

However, the health kiosks currently require *direct* interaction between users and the sensor technologies to apply them for measurement (e.g., finger clip, blood pressure cuff, chest strap, ECG, etc.). This reduces their acceptance (especially in older patients) and carries a risk of cross-contamination if the contact surfaces of the sensors are not cleaned regularly and adequately. Also, the correct application of the sensors partially requires expert knowledge. This can lead to measurement errors and therefore a lower quality of the data collected in a contact-based health kiosk.

Therefore, the objective of our research is the investigation, development and evaluation of non-contact robot-supported sensors as part of a health kiosk. To guide the users (patients in waiting rooms, clinics, emergency facilities, residents in senior residents) interactively through the (contactless) measurement procedure, a so-called *social interface* [5] is applied. Furthermore, to automatically adapt the contactless sensors to the height, pose and orientation of the patients and constantly keep a line-of-sight, a *robotic arm* is used.

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2 Health Kiosks

Health kiosks are increasingly deployed in public settings like supermarkets and pharmacies to measure vital signs such as heart rate, respiratory rate, and skin temperature without physical contact. A review of health kiosk technologies reveals that most systems focus on cardiovascular screening, with blood pressure being the most commonly measured parameter [6]. These kiosks integrate AI for real-time data analysis and decision support. Some include robotic arms to guide user interaction or dispense items [7]. One example for a robot is the ‘Dr. Spot’ platform [8] using IR and RGB cameras on a mobile quadruped robot for vital sign assessment while maintaining social distancing.

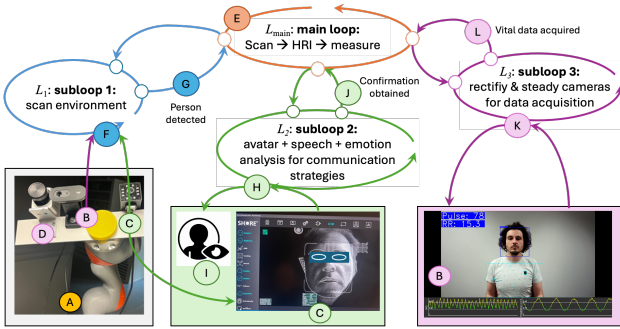


Figure 1: Overview of the hardware setup and software components. (A) robot arm, (B-D) cameras, (E) main loop L_{main} , (F, G) subloop L_1 for environment scan, (H-J) subloop L_2 for HRI, (K, L) subloop L_3 for contactless vital sign acquisition.

3 Methods

Figure 1 gives an overview of our currently used hardware setup and provides information about the main control loops, the subloops, and the social human-robot-interaction (HRI) scheme. A robotic arm (Kuka iiwa) (A) is carrying a set of different optical sensors, such as an RGB camera (B), an infrared (IR) camera (C) and a thermography camera (D). — The main loop L_{main} (E) initiates a subloop L_1 (F) to continuously scan the environment (‘detection mode’) using the robot-mounted cameras (B, C) for the presence of individuals in its vicinity [5] (Sec. 3.2). These individuals can either be standing or sitting in front of the system, including sitting in a wheelchair, or – in the next development stage – could be lying on a gurney. — If a person is detected (G), the main loop L_{main} initiates a subloop L_2 for HRI (‘interaction mode’) by seeking attention through multimodal social communication (H), including speech, visual signals [6], and (possibly) an avatar (I) mounted on the table (Sec. 3.3). — Once the person in the vicinity agrees to the assessment of vital signs (J), the main loop

proceeds with the contact-less measurement in subloop L_3 (K), (‘measurement mode’, Sec. 3.4), while constantly keeping a *line-of-sight* between the cameras and the area-of-measurement, namely the forehead and cheeks of the patient, using the robot arm. After storing the obtained vital data (L), the system returns to the main loop L_{main} (E).

3.1 Robot Control

Within each of the three subloops (L_1 , L_2 , L_3) the robot is related to a different task. (1) Within subloop L_1 (‘detection mode’, Sec. 3.2), the robot moves the flange (the tip) with the cameras 180° from left to right to scan for people in the vicinity. (In a later stage, also +/- 45° up and down movements are planned to detect patients on a gurney). — (2) Once a person is detected, within subloop L_2 the robot arm directs the IR-camera at a discrete distance to the person for basic emotion detection as part of social HRI scheme (‘interaction mode’, Sec. 3.3). Also, while the person remains within the close vicinity of the setup, the robot tries to keep a stable frontal view of the person with slow movements. — (3) Finally, within subloop L_3 , which focuses on the contactless acquisition of vital signs (‘measurement mode’, Sec. 3.4), the robot is used to orient and stabilize the RGB-camera for a constant line-of-sight towards the face (pulse) and torso (respiration). Even though it can be assumed that the person remains still during the measurement, the contactless data assessment is prone to ego-motions.

3.2 L_1 : Detection Mode

The main loop L_{main} (Fig. 1, E) gives control to subloop L_1 (‘detection mode’) to start with event detection, using situational awareness. By applying the visual language model Llava-01 [9], within subloop L_1 , the system scans and analyzes multimodal environmental data, hereby using visual and auditory input to identify a person entering the robot’s vicinity. Once a person has been detected within its vicinity and remains there, the acquired images are discarded, and the system shifts control to subloop L_2 (‘interaction mode’).

3.3 L_2 : Interaction Mode

Upon detection, the system uses subloop L_2 (‘interaction mode’) to assess the person’s emotional state [10], looking for signs of distraction, or readiness for interaction to initiate helpful communication proactively, see Fig. 1 (H-J). To this end, the software architecture leverages advanced cognitive and sensory capabilities to support a context-aware, proactive, and personalized dialogue. The design ensures that the subloop L_2

can guide people through the assessment of vital signs (L_3 , Sec. 3.4) while remaining sensitive to their readiness and focus.

Once a person has been detected (Sec.3.2) and their availability and willingness for interaction is confirmed, by making use of intention predictions based on proxemics [11] and gaze [12], the system activates its memory system to manage the interaction effectively: (1) *Short-term memory* tracks the ongoing context of the interaction, ensuring dialogue coherence and relevance. For instance, it keeps track of the steps already completed during the vital sign assessment (Sec. 3.3). (2) *Long-term memory* (LTM) can provide historical data, e.g., including prior interactions, individual preferences, and domain knowledge. This may enable the robot to tailor its communication to the person. However, LTM must be always implemented in tight collaboration with the data protection law and possible user consent.

The **dialogue generation module** plays a central role in guiding a person through the vital signs assessment. It employs a structured 3-step approach: (1) **Proactive Interaction:** Before initiating the communication, the system assesses the person's focus and availability to minimize distractions. If the person appears engaged and available, the subloop L_2 begins guiding them step-by-step through the contactless vital sign assessment in subloop L_3 . E.g., it may instruct the person to hold still, look at the camera for 10 to 15 seconds, or adjust their position for optimal line of sight. This proactive guidance ensures that the process is efficient and user-friendly. For training adequate proactive behavior, i.e., to decide the appropriate level of proactivity [13], reinforcement learning from human feedback [14] or reflection prompting approaches [15] are utilized. — (2) **Strategy Selection:** Based on acquired real-time data, individual context, and emotional state, the system selects appropriate communication strategies. If the person appears confused or hesitant, the system might slow down its instructions, provide additional explanations, or use a reassuring tone to build confidence. — (3) **Response Generation:** Based on the model's results, subloop L_2 generates clear, context-sensitive responses tailored to the task at hand. E.g., it may confirm successful data acquisition or guide the person to correct errors, such as adjusting their posture or ensuring proper lighting.

Throughout the process, the system employs *Retrieval-Augmented Generation* (RAG) [16] to enhance responses with accurate domain-specific information. For instance, it might explain the purpose of specific steps to the person, fostering trust and understanding. As the interaction proceeds, the system continuously monitors the person's emotional state (but discards the acquired images) and vital signs within subloops L_2 and L_3 . The system dynamically adapts its guidance if any issues arise – e.g., interruptions or suboptimal data quality. Upon completion of the health assessment (L_3) and consent of

the user, relevant data, including conversational context and vital records, are stored in the system's memory or can be sent to the electronic patient file. This ensures continuity and personalization in future interactions.

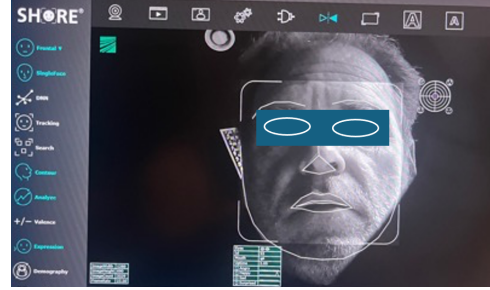


Figure 2: Capturing a person's emotional state (e.g. 'happy', 'sad', 'neutral', 'surprised') with an IR camera to obtain their availability and willingness for interaction.

3.4 L_3 : Measurement mode

For the contactless assessment of dynamic vital sign within the second subloop L_3 ('measurement mode'. Fig. 1.K), three image processing modules are used, namely for (a) the detection of the face, eyes and the chest, (b) the acquisition of the heartrate in the face and (c) the assessment of the respiration rate on the chest. For image acquisition, all RGB webcams with internal or external USB ports are supported. For our experiments we used a full-HD webcam with a frame rate of 30 FPS. — The contact-less **face detection** and localization is done by Haar-Wavelet based cascaded classifier by Viola & Jones [17], which provides real-time information about the face in the scene with a bounding box B_{face} of height (h) and width (w) as well as regions and location L_{eyes} corresponding to the eyes. The approach for the assessment of the **respiration rate** is inspired by Gwak et al. [18]. Using the location of the facial mask B_{face} , the chest is located by generating a bounding box B_{chest} $h/4$ below B_{face} and of twice the width w and $3/4$ of the height h , see Fig. 3(B). Inside the chest area B_{chest} corner features f are located using the Shi & Tomasi feature detector [19], which are then tracked using the LKT optical flow approach running in real time on almost any hardware [20, 21]. The movement of the features are stored in a $t = 15$ second sliding window and a DFT to obtain the frequency spectrum of the chest movement, induced by breathing. The highest peak in the power spectrum represents the estimated respiration rate. — Measuring the **heart rate** via RGB camera is based on extracting the photoplethysmogram (PPG) signal and approximating the heart rate [22]. We use an extension of a particle filter employing a generative model to simulate the measurement process, modeling HR and the PPG signal indirectly via an Ornstein-Uhlenbeck process (OU-process),

which accounts for motion and illumination changes. For the assessment of the heart rate, we use RGB data from three patches inside the bounding box B_{face} . As depicted in Fig. 3(A), the first patch p_1 with height $h_{p1} = h/8$, and the lower edge is located $h/8$ above the center point L_{eyes} between the eyes. Patches p_2 and p_3 are located $h/8$ below the eyes. Each patch measures $h/8 \times h/8$. — Fig. 4 provides an example of the extracted pulse (left) and respiratory rate (right).

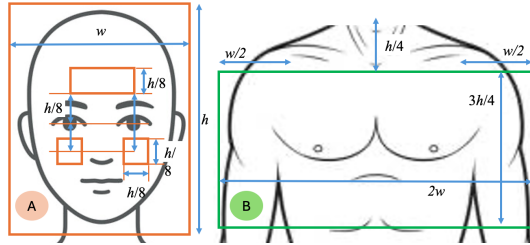


Figure 3: (A) Placement of measurement patches for the heart rate w.r.t. to the bounding boxes of the face and location of the eyes. (B) bounding box on the chest for corner detection.

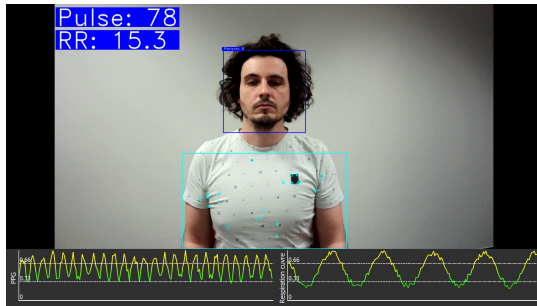


Figure 4: Example of pulse (left) and respiratory rate (right).

4 Discussion

In this concept paper, we have introduced a robotic supported health kiosk that can detect people in its vicinity, engage in a socially attuned and proactive dialogue, and guide to the contactless assessment of vital signs. By combining a detection with an interaction mode focussing on socially aware guidance, the proposed health kiosk considers the importance of user-friendly design for technical systems.

The concept shows how a health kiosk can sidestep the hygiene and measurement issues associated with previous stations. So, it is laying groundwork for feasible alternatives to today's health-kiosk by offering a contactless assessment of vital signs. The system's modular architecture makes it scalable and allows for an easy integration of sensors to record further vital signs. In addition, there is no need to store raw data. If the storage of data in the LTM or electronic patient file is desired, it can store processed metrics required for a recurring interaction between the robotic kiosk and an individual, hence following current data safety concepts and GDPR, that would have to be included.

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